

The Clarkson–Kruskal direct method on two examples: the heat kernel and the hydrogen J_0 solution

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A companion note to *A New Method of Solving PDEs*. The Clarkson–Kruskal (CK) *direct method* [1, 2] finds similarity reductions of a PDE — ways to collapse it onto an ODE — by substituting a structured ansatz and forcing the result to be an ODE, with no use of group theory. We work it on two of the paper’s own example PDEs: the linear heat equation $u_t = u_{xx}$ and the hydrogen Schrödinger equation $-\frac{1}{2}\nabla^2\Psi - \frac{1}{r}\Psi = E\Psi$.

1 The recipe

For a PDE in two independent variables, seek

$$u(x, t) = \alpha(x, t) + \beta(x, t) w(z(x, t)), \quad (1)$$

with w an unknown function of the single *similarity variable* z , and α, β, z to be determined. Substituting into the PDE produces a sum $\sum_i \Gamma_i(x, t) W_i(w)$, where each W_i is a derivative or power of w . For this to be an ODE in z , every coefficient must equal a chosen normalizer Γ_0 times a *pure function of z* ,

$$\Gamma_i(x, t) = \Gamma_0(x, t) \theta_i(z(x, t)), \quad (2)$$

giving an overdetermined system (the *determining equations*) for α, β, z together with the reduced ODE for w . The representation (1) is redundant under $z \mapsto Z(z)$, $\beta \mapsto \beta/\Omega(z)$, and $\alpha \mapsto \alpha + \beta\Omega(z)$; CK fix these three freedoms to make the system solvable.

2 Example 1: the heat equation $u_t = u_{xx}$

Writing u for Ψ , chain-rule (1) through the equation:

$$u_t = \alpha_t + \beta_t w + \beta z_t w', \quad u_{xx} = \alpha_{xx} + \beta_{xx} w + (2\beta_x z_x + \beta z_{xx}) w' + \beta z_x^2 w''.$$

So $u_t - u_{xx} = 0$ groups, by the structure in w , into

$$\underbrace{-\beta z_x^2}_{w''} w'' + \underbrace{(\beta z_t - 2\beta_x z_x - \beta z_{xx})}_{w'} w' + \underbrace{(\beta_t - \beta_{xx})}_w w + \underbrace{(\alpha_t - \alpha_{xx})}_1 = 0. \quad (3)$$

Dividing by the normalizer $\Gamma_0 = -\beta z_x^2$ and imposing (2) gives the determining equations

$$\beta z_t - 2\beta_x z_x - \beta z_{xx} = -\beta z_x^2 \theta_1(z), \quad (4)$$

$$\beta_t - \beta_{xx} = -\beta z_x^2 \theta_2(z), \quad (5)$$

$$\alpha_t - \alpha_{xx} = -\beta z_x^2 \theta_3(z), \quad (6)$$

with reduced ODE $w'' + \theta_1(z) w' + \theta_2(z) w + \theta_3(z) = 0$.

Normal form. Using the freedoms, take the constant-coefficient linear normal form $\alpha = 0$, $z = a(t)x + b(t)$, $\beta = \beta(t)$. Then (6) is satisfied, and (4)–(5) reduce to

$$a'x + b' = -a^2 \theta_1(z), \quad \beta'/\beta = -a^2 \theta_2(z).$$

The right-hand side of the first is affine in x only if $\theta_1(z) = pz + q$, forcing $a' = -pa^3$, $b' = -a^2(pb + q)$; the second forces $\theta_2 \equiv c$, a constant.

Two branches. The choice $p = 0$ gives a, b constant and $\beta = e^{-ca^2t}$: these are the *separable* Fourier modes $u = e^{-ck^2t}e^{ikx}$, i.e. ordinary separation of variables. The choice $p \neq 0$ is the genuinely non-separable branch. Solving $a' = -pa^3$ with $p = \frac{1}{2}$, time origin 0, and $q = 0$ (so $b \equiv 0$):

$$a^2 = \frac{1}{t}, \quad z = \frac{x}{\sqrt{t}}, \quad \beta = t^{-c}.$$

With $\theta_1 = \frac{1}{2}z$ and $\theta_2 = c$ the reduced ODE is the one-parameter family

$$w'' + \frac{1}{2}zw' + cw = 0, \tag{7}$$

the free c being the scaling weight. The distinguished member $c = \frac{1}{2}$ has the closed-form solution $w(z) = e^{-z^2/4}$ (indeed $w'' = (-\frac{1}{2} + \frac{z^2}{4})w$, $\frac{z}{2}w' = -\frac{z^2}{4}w$, and these plus $\frac{1}{2}w$ cancel). Reassembling $u = \beta w(z)$,

$$\boxed{u(x, t) = \frac{1}{\sqrt{t}} e^{-x^2/4t}} \tag{8}$$

— the heat kernel, exactly the solution quoted in the paper. The direct method has *derived* it, and exhibited the separable Fourier modes as the sibling branch of the same determining system.

3 Example 2: hydrogen, and the J_0 solution

The paper's marquee solution is $\Psi = J_0(2\sqrt{x+r})$ with $r = \sqrt{x^2 + y^2 + z^2}$, a function of the single *parabolic* coordinate

$$\xi = x + r. \tag{9}$$

This is the natural similarity variable: take the pure reduction ansatz $\Psi = w(\xi)$ (i.e. $\alpha = 0$, $\beta = 1$, $z = \xi$ in (1)) and ask the CK question — when does the equation close into an ODE in ξ ?

The reduced operator. With $\nabla\xi = \hat{e}_x + \hat{r}$ we get the two field quantities

$$|\nabla\xi|^2 = 2 + 2\frac{x}{r} = \frac{2\xi}{r}, \quad \nabla^2\xi = \nabla^2r = \frac{2}{r},$$

so that $\nabla^2\Psi = |\nabla\xi|^2w'' + (\nabla^2\xi)w' = \frac{2}{r}(\xi w'' + w')$. The Schrödinger equation $-\frac{1}{2}\nabla^2\Psi - \frac{1}{r}\Psi = E\Psi$ becomes, after multiplying by $-r$,

$$\xi w'' + w' + w = -Erw. \tag{10}$$

Closure forces $E = 0$. Read (10) through the CK lens. With normalizer $\Gamma_0 = -\frac{1}{2}|\nabla\xi|^2 = -\xi/r$, the w' coefficient ratio is $\theta_1 = \frac{-1/r}{-\xi/r} = \frac{1}{\xi}$, a function of ξ — good. But the w coefficient ratio is

$$\theta_2 = \frac{-1/r - E}{-\xi/r} = \frac{1 + Er}{\xi},$$

which is a function of ξ *only if* Er is. Since $r = \frac{1}{2}(\xi + \eta)$ in parabolic coordinates ($\eta = r - x$), r is independent of ξ , so the closure condition (2) holds **iff** $E = 0$. The direct method's own consistency requirement selects the zero-energy sheet — precisely where the paper's solution lives.

The reduced ODE. At $E = 0$, (10) is

$$\xi w'' + w' + w = 0. \tag{11}$$

The substitution $\tau = 2\sqrt{\xi}$ (so $\xi = \tau^2/4$) turns (11) into Bessel's equation of order zero,

$$W_{\tau\tau} + \frac{1}{\tau}W_{\tau} + W = 0, \quad W(\tau) = J_0(\tau),$$

giving

$$\boxed{\Psi = J_0(2\sqrt{x+r})}. \tag{12}$$

Both the reduction and the final wavefunction were checked symbolically: $J_0(2\sqrt{\xi})$ satisfies (11) identically, and $J_0(2\sqrt{x+r})$ annihilates $-\frac{1}{2}\nabla^2\Psi - \frac{1}{r}\Psi$ to machine precision at generic points.

4 Reading against the new method

On both examples the direct method *forces a single similarity variable into existence* — $z = x/\sqrt{t}$ for heat, $\xi = x + r$ for hydrogen — and then reads off whatever ODE in that variable is consistent. It is organized around the equation's (implicit) symmetries: the heat run even hands back the separable Fourier modes and the non-separable heat kernel as two arms of one determining system, and the hydrogen run lets the closure condition itself pick out $E = 0$.

The new method is organized differently. It does not posit (1); it fixes a *differential-polynomial* ansatz, adjoins the PDE, reduces, and prime-decomposes, certifying via the completeness theorem that every solution of that ansatz form has been found. The two routes meet at the same wavefunctions — the heat kernel and the hydrogen J_0 — from opposite organizing principles: the direct method supplies the reduction but no completeness claim; the new method supplies the guarantee but must be handed the ansatz as differential polynomials.

References

- [1] P.A. Clarkson, M.D. Kruskal, New similarity reductions of the Boussinesq equation, *J. Math. Phys.* **30** (10) (1989) 2201–2213.
- [2] P.A. Clarkson, E.L. Mansfield, Symmetry reductions and exact solutions of a class of nonlinear heat equations, *Phys. D* **70** (3–4) (1994) 250–288.